Updates on Randomness on LLM And Internship Final Report

Project Proposal, JAIST Summer 2025 Research Internship
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2025/07/31

Updates from the past 2 weeks

- Read more papers on LLM bias and hallucinations
- Tried using SAE and Circuit Tracing using Neuronpedia
  - Extract features
  - Intervene the features
- Tried t-SNE in feature latent space

Papers

Read

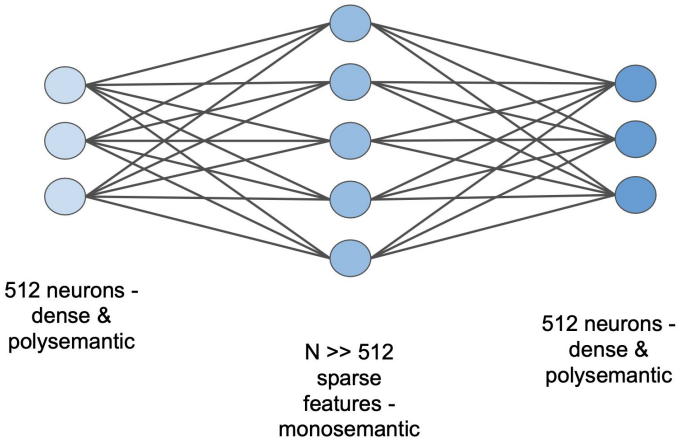
- Surveys on hallucinations
- Hallucinations prevention techniques

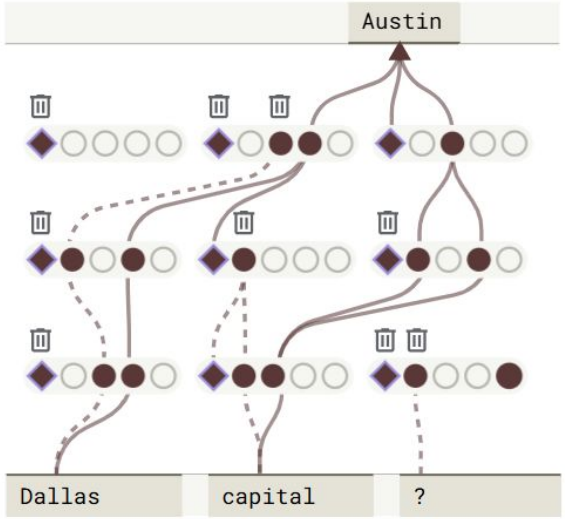
Result

- No new paper related to number/word randomness concept (except the Benford's Curse paper)
- Only randomness on the decoder architecture (temperature, top-K, etc.)

Neuronpedia

A website for interacting with trained **SAEs and Anthropic's Circuit Tracing models**



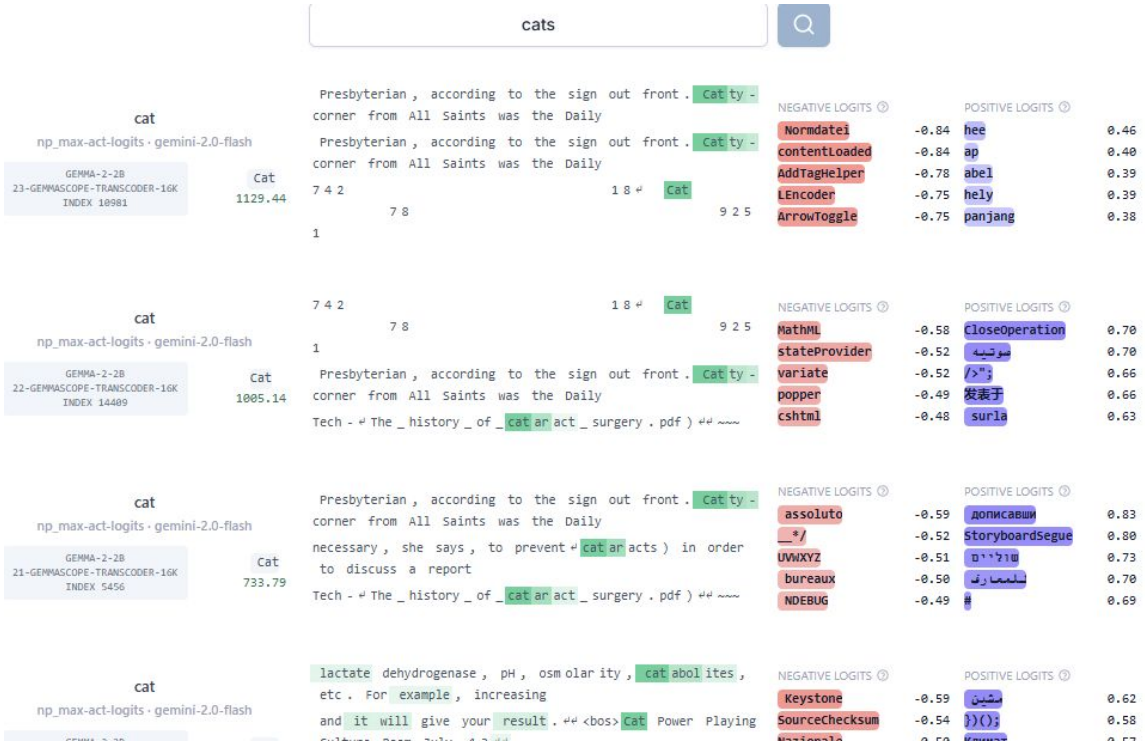


Neuronpedia

SAE example

Neurons related to “cats”

[Link]



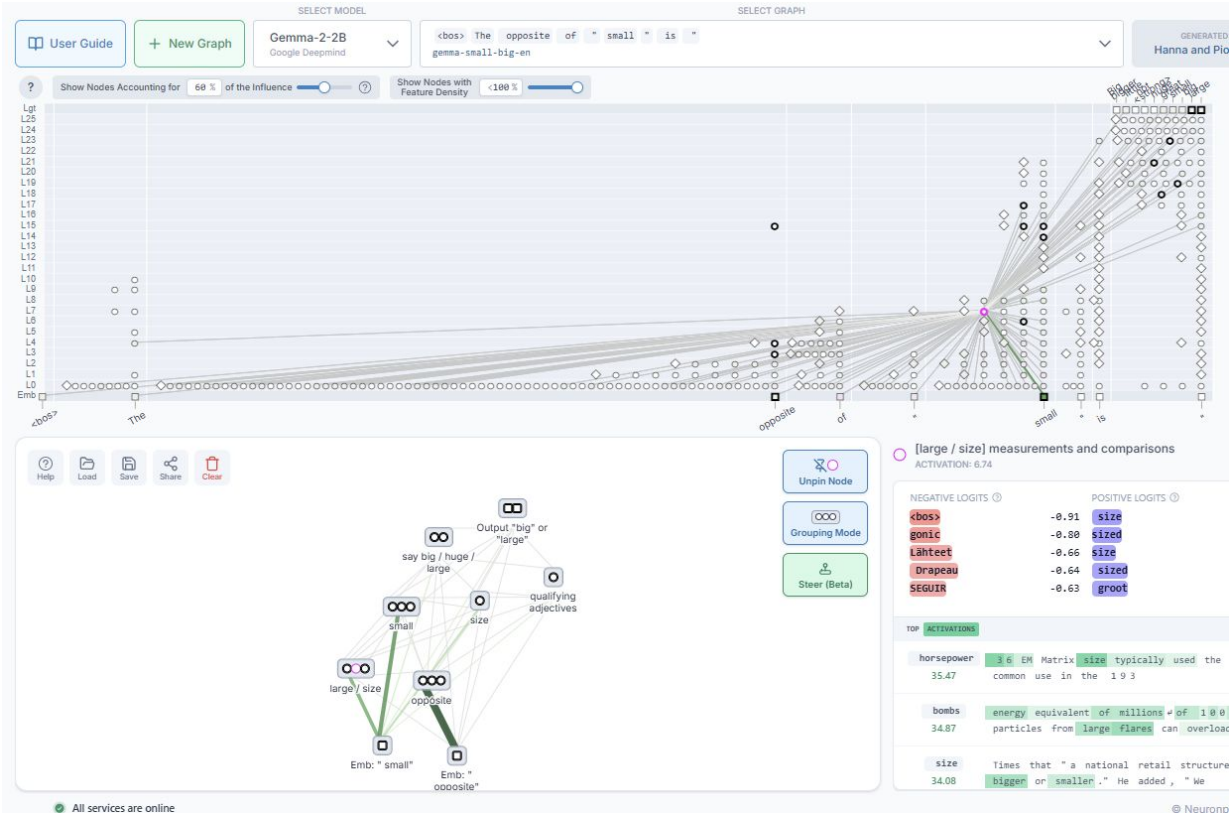
Neuronpedia

Circuit Tracing example

Neurons connections from the prompt

The opposite of "small" is "

[Link]



Randomness

Tries some prompts about numbers and words randomization

- Random
- Random from a numbers/words list
- Random but with bias

Did

- Group the nodes
- Steer the neurons

SAE

What is 1+1? Answer in one word

Result

- Default => 2
- Reduce “two” features Link => weird random tokens

SAE

Please random a number. Answer in one word

Result

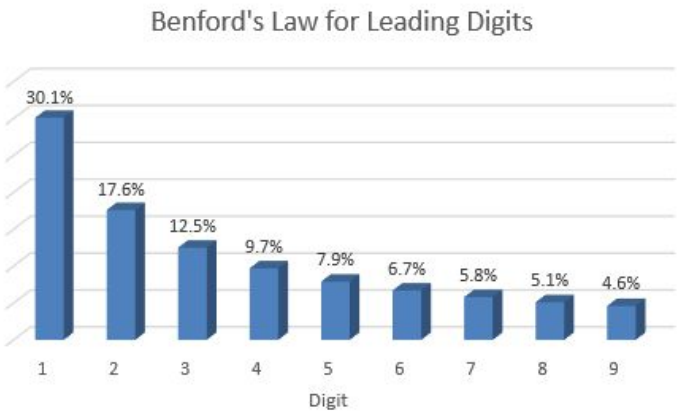
- Default => 42
- Increase “one” features Link => 7

SAE

Based on Benford's Law, what is the most used number? Answer in one word

Result

- Default => One
- Reduce “one” features Link => Numbers
- Increase “two” features Link => One.



Circuit Tracing

A random number

Result

- Graph Link
- Smaller numbers => higher Probs

Circuit Tracing

Please random a number between 1 and 10

The number is "

Result

- Graph Link
- Token “1” => 0.15
- Then rest => less than 0.09

Circuit Tracing

A random number from the list of 1, 5, and 7 is

Result

- Graph + some grouping Link
- Probs of 5 > 1 > 7
- Around 0.2x Not 0.33

Circuit Tracing

Random a number from the list of 57, 301, 975, 1884 is

Result

- Graph Link
- Token “1” => 0.29 Prob

Circuit Tracing

The random word related to technology is "

Result

- Graph Link
- Distributed small probs
- A lot of possible tokens

Circuit Tracing

Please randomize a word from "Cactus", "Satellite", "Bangkok", "Pancake", "Justice"

The word is: "

Result

- Graph Link
- Similar probs

Circuit Tracing

According to Benford's Law, the most used number between 1 and 9 is

Result

- Graph Link
- Token “1” => 0.92 Prob
- Couldn’t find “Benford” token (just “Ben” and “Ford” features)

Circuit Tracing

Benford number is

Result

- Graph Link
- Token “1” => 0.81 Prob
- Couldn’t find “Benford” token (just “Ben” and “Ford” features)

Conclusion

Most of the cases, the final token probabilities make sense

SAEs

- Intervene 1-2 features => didn't really change or mean anything
- Might select more related features

Circuit Tracing

- Grouping related features => didn't really change
- Need to explore more features

Note: Manually interpreting the features is very hard

More ideas to do (mainly for Circuit Tracing)

Try more cases

- Rank things or evaluate some scores (LLM as a judge)

Automatic Features Labelling/Classification

- Latent Space Clustering (PCA/t-SNE/UMAP/etc.)
- Mech Interp on Circuit Tracing?
- Formulate as a new NLP task (New Field?)

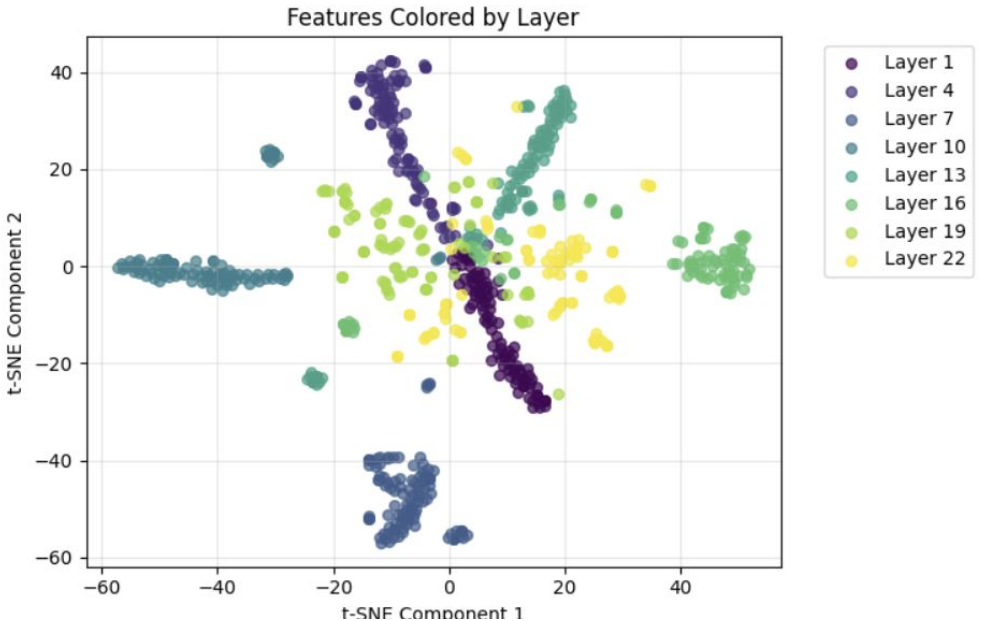
More Intervention

- Delete the whole feature (not just reduce)
- Combine features?

Current Results

Some t-SNE from the Circuit Tracing model

(Not enough time to explore more)



Final Report

Week 1

- Onboarding
- Meeting on future topics(Spatial Benchmark or AI Safety)
- Read more AI Safety papers

Week 2

- Read more AI Safety papers

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Week 3

- Read more papers
- Meeting on Hidden Chains of Thought (define/read/try find cases)

Week 4

- Explore datasets
- Start evaluating CoT faithfulness using simple math and thai exam dataset

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Week 5

- Evaluate reasoning games (20 questions/Thai Tricks games)
- Explore LLM randomization from 20 questions
- Read papers on Stateless and Multi-turn

Week 6

- Explore poem writing, logic problems, numbers/words randomization

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Week 7

- Test randomness using Benford/Zipf's Laws
- Read more about human psychology on randomness
- Read more about debiasing, causal, and mech interp

Week 8

- **This presentation**
- Use Neuronpedia's SAE and Circuit Tracing for randomness prompts
- Presented in the reading group about "On the Biology of a Large Language Model" paper by Anthropic

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Week 9

- **This presentation**
- Dig deeper on Circuit Tracing code/weights to analyse latent vectors

Things to do next

Future plans

- Chula's Final Project?
- Future of this LLM randomness project

Reading Materials

- Some nice paper?

Next meeting

- Next month? After going back to Thailand (what to continue on)